

2(a)	Line 1 should be Counter ← 0 Line 3 RandNum[Counter] ← Rand(1, 100) should be RandNum[Counter] ← Rand(1, 101) Line 4 Counter ← Counter + 2 should be Counter ← Counter + 1 Line 5 UNTIL Count ≤ 50 should be UNTIL Counter ≥ 50 // UNTIL Counter = 50 <pre> 1 Counter ← 0 2 REPEAT 3 RandNum[Counter] ← Rand(1, 100) 4 Counter ← Counter + 1 5 UNTIL Counter ≥ 50 Or Line 3 RandNum[Counter] should be RandNum[Count] Line 3 Rand(1, 100) should be Rand(1, 101) Line 4 Counter ← Counter + 2 should be Count ← Count + 1 Line 5 UNTIL Count ≤ 50 should be UNTIL Count ≥ 50 // UNTIL Count = 50 1 Count ← 0 2 REPEAT 3 RandNum[Count] ← Rand(1, 100) 4 Count ← Count + 1 5 UNTIL Count ≥ 50 </pre>	4
2(b)	One mark for each correct line <pre> FOR Count ← 0 TO 49 // FOR Count ← 1 TO 50 RandNum[Count] ← Rand(1, 101) / Rand(0, 101) NEXT // NEXT Count </pre>	3
2(c)	Precondition loop // WHILE ... DO ... ENDWHILE	1

Question	Answer	Marks
4(a)	One mark for error identified and suggested correction (Max three) Line 8 OUTPUT Value2 – should be INPUT Value2 Line 9 IF Operator – should be CASE OF Operator Line 15 OUTPUT "The answer is ", Value1 – should be Answer The loop may be corrected using a number of alternative methods: One mark for error identified and suggested correction (Max two) Method 1 Line 1 Continue ← 1 should be Continue ← 0 Line 22 UNTIL Continue = 0 should be ENDWHILE // Line 2 WHILE Continue = 0 should be REPEAT and Line 22 UNTIL Continue = 0 should be Until Continue = 1 OR Method 2 Line 2 WHILE Continue = 0 should be REPEAT Line 20 Continue ← 1 should be Continue ← 0 // Line 1 Continue ← 1 should be Continue ← 0 and Line 22 UNTIL Continue = 0 should be Until Continue = 1 OR Method 3 Line 2 WHILE Continue = 0 should be WHILE Continue = 1 Line 20 Continue ← 1 should be Continue ← 0 and Line 22 UNTIL Continue = 0 should be ENDWHILE	5

Question	Answer	Marks
4(a)	Corrected algorithm example 1 <pre> 1 Continue ← 0 2 WHILE Continue = 0 (DO) 3 OUTPUT "Enter 1 for +, 2 for -, 3 for * or 4 for /" 4 INPUT Operator 5 OUTPUT "Enter the first value" 6 INPUT Value1 7 OUTPUT "Enter the second value" 8 INPUT Value2 9 CASE OF Operator 10 1: Answer ← Value1 + Value2 11 2: Answer ← Value1 - Value2 12 3: Answer ← Value1 * Value2 13 4: Answer ← Value1 / Value2 14 ENDCASE 15 OUTPUT "The answer is ", Answer 16 OUTPUT "Do you wish to enter more values (Yes or No)?" 17 INPUT MoreValues 18 IF MoreValues = "No" 19 THEN 20 Continue ← 1 21 ENDIF 22 ENDWHILE </pre>	

Question	Answer	Marks
4(a)	Corrected algorithm example 2 <pre> 1 Continue ← 1 2 REPEAT 3 OUTPUT "Enter 1 for +, 2 for -, 3 for * or 4 for /" 4 INPUT Operator 5 OUTPUT "Enter the first value" 6 INPUT Value1 7 OUTPUT "Enter the second value" 8 INPUT Value2 9 CASE OF Operator 10 1: Answer ← Value1 + Value2 11 2: Answer ← Value1 - Value2 12 3: Answer ← Value1 * Value2 13 4: Answer ← Value1 / Value2 14 ENDCASE 15 OUTPUT "The answer is ", Answer 16 OUTPUT "Do you wish to enter more values (Yes or No)?" 17 INPUT MoreValues 18 IF MoreValues = "No" 19 THEN 20 Continue ← 0 21 ENDIF 22 UNTIL Continue = 0 </pre>	

Question	Answer	Marks
4(b)	One mark per bullet MP1 Appropriate loop (begin and end) / otherwise selection MP2 Testing both ends of condition MP3 Suitable message MP4 Input/re-input WHILE Operator < 1 OR Operator > 4 (DO) <pre> OUTPUT "Enter 1, 2, 3 or 4" INPUT Operator ENDWHILE </pre> Alternative answer <pre> REPEAT IF Operator < 1 OR Operator > 4 THEN OUTPUT "Enter 1, 2, 3 or 4" INPUT Operator ENDIF UNTIL Operator >= 1 AND Operator <= 4 </pre> One mark After line 4 / between lines 2 and 5	5

NOVEMBER 2021 (3)

Question	Answer	Marks
Section B		
2(a)	Line 1 should be Counter Line 2 should be < 50 Line 3 should be RandUp(0,100) // RandUp(-1,100) Line 4 should be Counter ← Counter + 1 1 Counter ← 0 2 WHILE Counter < 50 DO 3 NumRand[Counter] ← RandUp(0, 100) 4 Counter ← Counter + 1 5 ENDWHILE	4
2(b)	Any three from: FOR Counter ← 0 TO 49 // FOR Counter ← 1 TO 50 NumRand[Counter] ← RandUp(0,100) / RandUp(-1,100) NEXT // NEXT Counter	3

MARCH 2022

Question	Answer	Marks
Section B		
2(a)	One mark per mark point, max four • 100 • AND Age < 12 • Count12to18 + 1 • CountOver18	4
2(b)	One mark suitable IF construct, one mark correct assignment statement, for example IF Age < 7 THEN CountUnder7 ← CountUnder7 + 1 ENDIF One mark suitable message, one mark correct use of countUnder7 variable, for example OUTPUT "There are ", CountUnder7, " students aged under 7."	4

JUNE 2022 (1)

PUBLISHED

Question	Answer	Marks
4(b)	One mark per mark point, max four MP1 appropriate conditional loop structure MP2 correct identification of invalid input MP3 appropriate error message MP4 repeated input of score until correct WHILE Score < 0 OR Score > 100 (DO) OUTPUT "Your entry must be between 0 and 100, inclusive, please try again " INPUT Score ENDWHILE Or: REPEAT IF Score < 0 OR Score > 100 THEN OUTPUT "Your entry must be between 0 and 100, inclusive, please try again " INPUT Score ENDIF UNTIL Score >= 0 AND Score <= 100	4

Question	Answer	Marks
Section B		
2(a)	One mark per mark point, max three - line 8 / PassCheck ← TRUE correction PassCheck ← FALSE - line 12 / IF Password <> Password correction IF Password2 <> Password // IF Password <> Password2 - line 18 / UNTIL PassCheck OR Attempt <> 3 correction UNTIL PassCheck OR Attempt = 3 / UNTIL PassCheck OR Attempt >= 3	3
2(b)	One mark check, one mark matching description, max four Check: validation // length check Description length check // checks number of characters in password Check: verification // double entry Description double entry // comparison that two inputs are the same	4
2(c)	One mark per set, one mark matching reason, max four Set 1 – any appropriate example e.g. "small" Reason must follow through from the password given e.g. abnormal data will be rejected Set 2 – any different appropriate example e.g. "password" and "password" Reason must be different and follow through from the password given e.g. normal data will be accepted	4

Question	Answer	Marks
3(a)	One mark per mark point, max two - a list / one column of items ... of the same data type ... stored under a single identifier ... with a single index to identify each element One mark for an example of a declaration - example e.g. DECLARE MyArray[1:10] OF INTEGER	3

Question	Answer	Marks
3(b)	One mark per mark point, max two - using a counter to index the array - so that the same code can be repeatedly used to check every element // every element can be checked in a loop	2